Parallel circuits

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Relate branch currents to a shared potential difference.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 5.*
$Q=It=15$ C.

02. Worked example: explain the model
Same 6 V across branches; 1+2=3 A at each junction.

03. Read and interpret the model
Yes, if its own complete path to the ideal supply remains.

04. Apply the physics
2 A and 1 A; total 3 A.

05. Explain your reasoning
Adding a conducting path increases total current for the same voltage; $R_{eq}=V/I_{total}$ decreases.

06. Review: connect an earlier idea *Review: Unit 6, Day 14.*
A diverging lens spreads initially parallel rays outward.

07. Exit check
0.5 A.

08. STRETCH: test the model
Existing currents unchanged; new branch 1 A; total 4 A, assuming ideal maintained voltage.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 5.*
Circuit response involves propagation of electromagnetic changes and local charge motion. Drift speed is not the propagation speed of the signal.

02. Worked example: explain the model
Same 6 V across branches; 1+2=3 A at each junction.

03. Read and interpret the model
Yes, if its own complete path to the ideal supply remains.

04. Apply the physics
2 A, 1 A; total 3 A; R_{eq}=4 Ω.

05. Explain your reasoning
Adding a conducting path increases total current for the same voltage; $R_{eq}=V/I_{total}$ decreases.

06. Review: connect an earlier idea *Review: Unit 6, Day 14.*
d_i≈-13.33 cm; m≈+1.667. Upright, enlarged virtual image.

07. Exit check
4 Ω parallel; 16 Ω series.

08. STRETCH: test the model
R_{eq}=2.4 ohms; I=5 A; power 60 W, also 36+24 W.